DAX AND DATA VISUALIZATION

POWER BI -ASSIGNMENT 2

Overview:

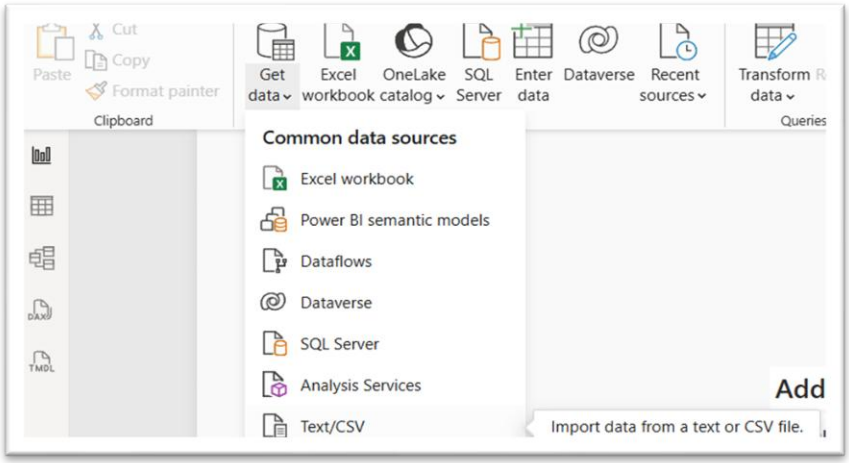
E-Commerce Sales Analysis project uses Power BI to analyse sales data, create key business metrics using DAX, and develop interactive dashboards to visualize sales performance, profitability, and target achievement for better decision-making

Process Flow:

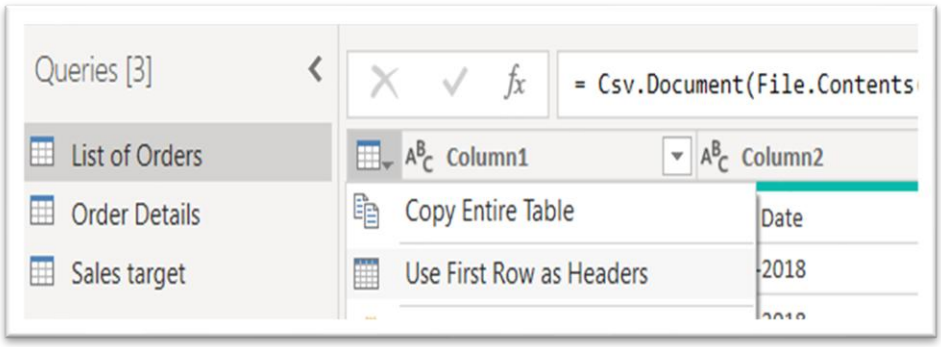
1. Import data
2. Clean and transform data
3. Creating relationships between tables
4. Creating calculated columns and DAX measures
5. Build visualizations and dashboard

Description:

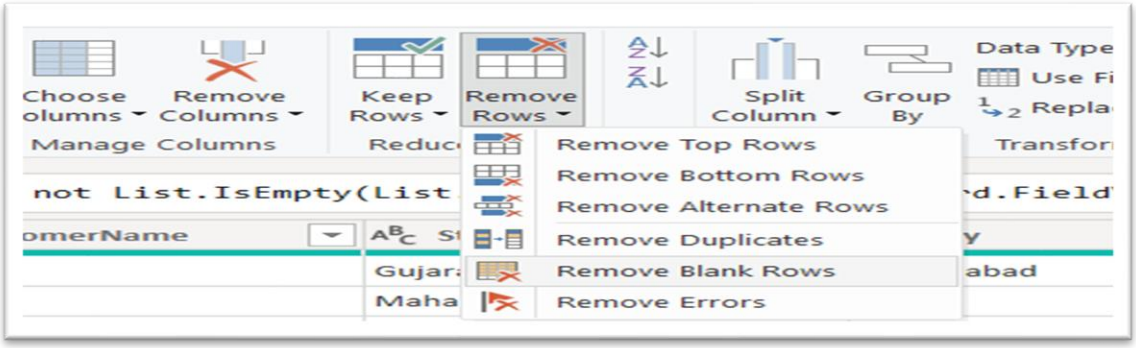
1. **Import data:**
   - ✓ Uploaded the Dataset files using the **Get Data->Text/CSV**
   - ✓ Gave **Transform** Option to Clean and Transform the data



2. **Clean and transform data:**
   List of Orders Table
   - ✓ Changed the First rows as columns



- ✓ Removed the Blank rows by **Remove Rows-> Remove Blank rows**
- ✓ Cross checked the Values by **Transform-> Statistics-> Count Distinct values and Count Values**

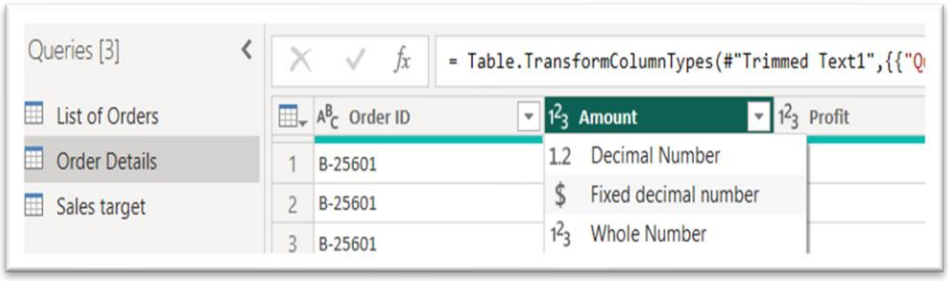


- ✓ Extracted the Month from the Order date by **Add column-> Date-> Name of the Month**

Order Details:

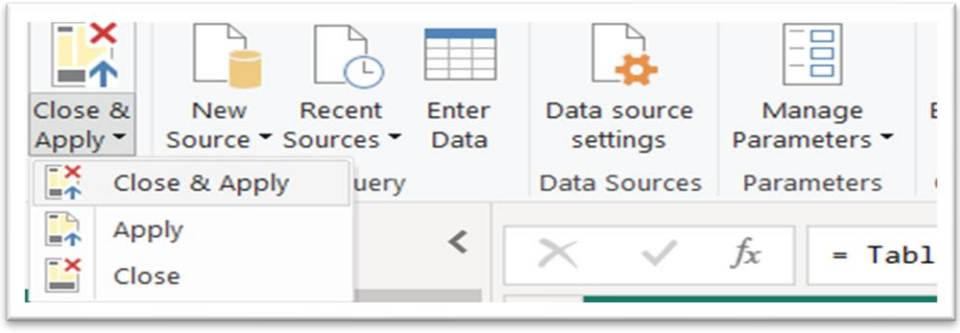
There are no null values, and blank rows so changed the Data type for,

- **Amount and Profit** - Fixed Decimal Number
- **Quantity**- Whole numbers
- Trimmed and Clean the **Category** column



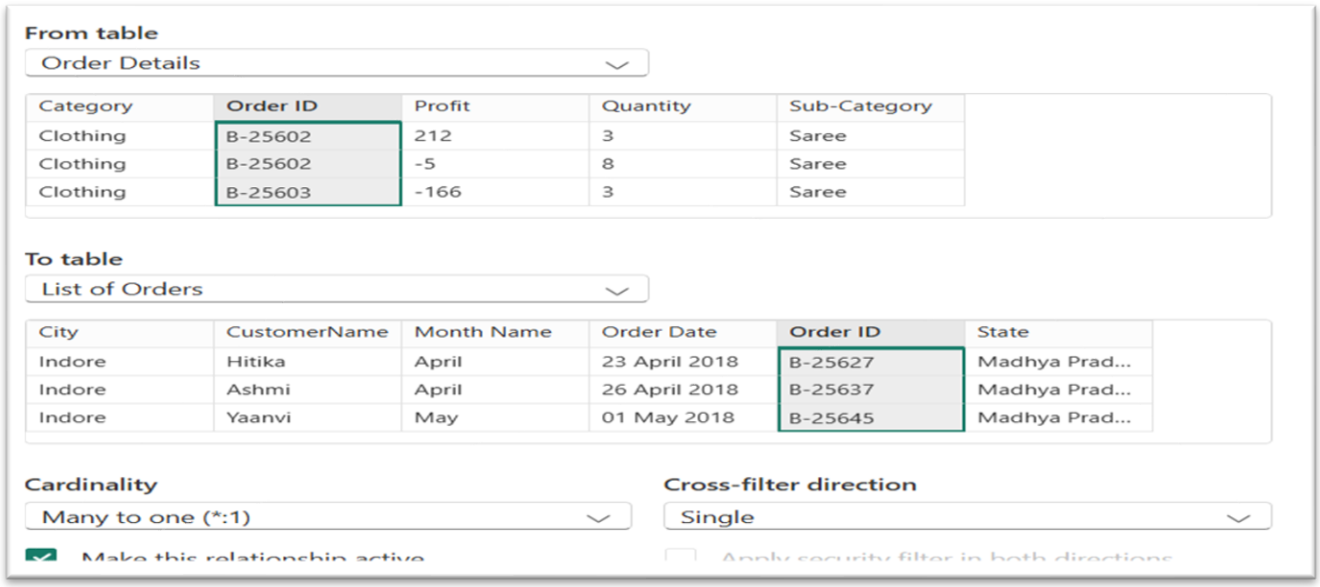
Sales Target:

- ✓ Extracted the Month from the Month of Order date by **Add column-> Date-> Name of the Month**
- ✓ Changed the Data type for **Target** Column to Fixed Decimal Number
- ✓ By choosing **Home-> Close & Apply** loaded the transformations into Power BI

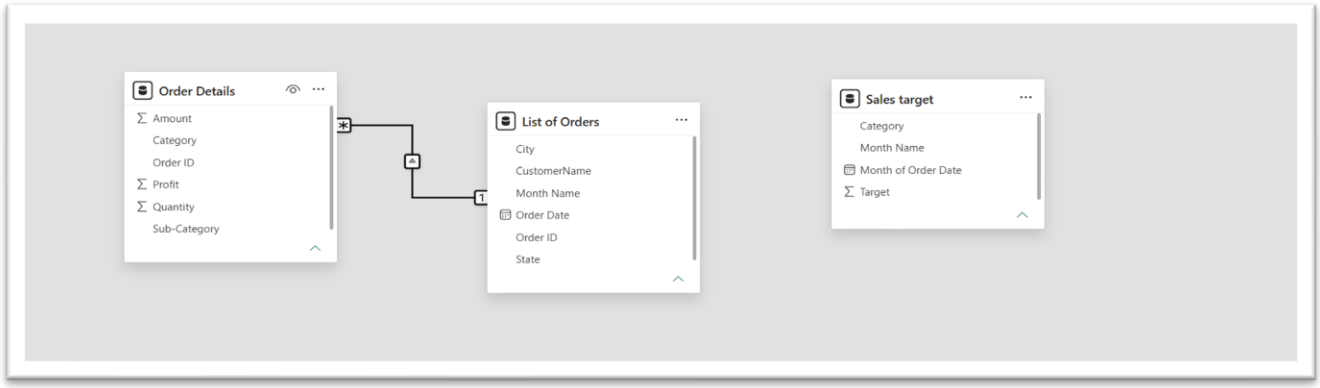


3. **Creating relationships between tables:**

- ***Data Modelling:***
  - ✓ By designating the **Order Details** as Fact table Created the Relation to **List of Orders** by Order ID as Primary Key
  - ✓ **Sales Target** won't be part because it has summary version



Initial Model:



4. **Creating calculated columns and DAX measures:**

- ***Calculated Columns:***

  Created a Calculated Column below Columns using DAX:
  - ✓ **Category Type** = 'Order Details'[Category]&"-"&'Order Details'[Sub-Category]
  - ✓ **Revenue** = 'Order Details'[Amount]*'Order Details'[Quantity]
  - ✓ **Sales Category** = SWITCH(TRUE(),'Order Details'[Amount]>=2000,"Expensive",'Order Details'[Amount]>=1000,"High",'Order Details'[Amount]>=500,"Medium","Low")

  Additional Column:
  - ✓ **Profit Status** = SWITCH(TRUE(),'Order Details'[Profit]>0,"Profitable",'Order Details'[Profit]=0,"Zero Profit", "Loss")

- ✓ **Order Size** = IF('Order Details'[Quantity]>10,"Bulk Order", IF('Order Details'[Quantity]>=5,"Regular Order", "Light Order"))

- ***Calculated Measures:***

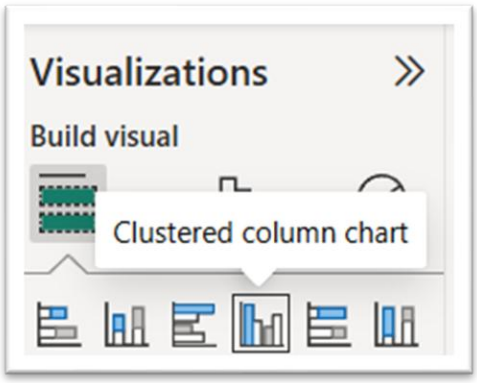
  Created a Measure for columns using DAX:
  - ✓ ***Total Orders*** *= COUNT('Order Details'[Order ID])*
  - ✓ ***Avg profit in Delhi*** *= CALCULATE(AVERAGE('Order Details'[Profit]),'List of Orders'[State]="Delhi")*
  - ✓ ***YTD Sales*** *= TOTALYTD(SUM('Order Details'[Amount]),'Calendar table'[Date])*
  - ✓ ***Total Sales*** *= SUM('Order Details'[Amount])*
  - ✓ ***Total Profit*** *= SUM('Order Details'[Profit])*
  - ✓ ***Sales Achievement %*** *= DIVIDE([Total Sales],SUM('Sales target'[Target]),0) * 100*
  - ✓ `Total Target = SUM('Sales target'[Target])`
  - ✓

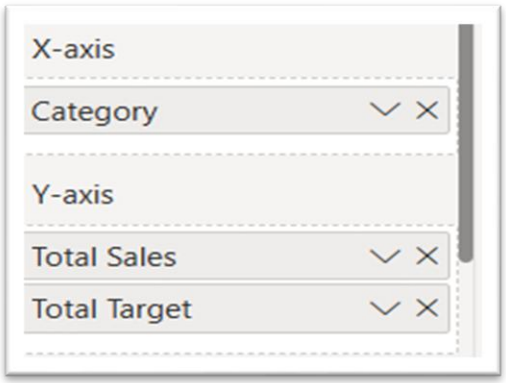
- ***Calculated Table:***

  Created a Calculated Table for relationship purpose using DAX:
  - ✓ ***Calender table*** = CALENDAR(MIN('List of Orders'[Order Date]),MAX('List of Orders'[Order Date]))

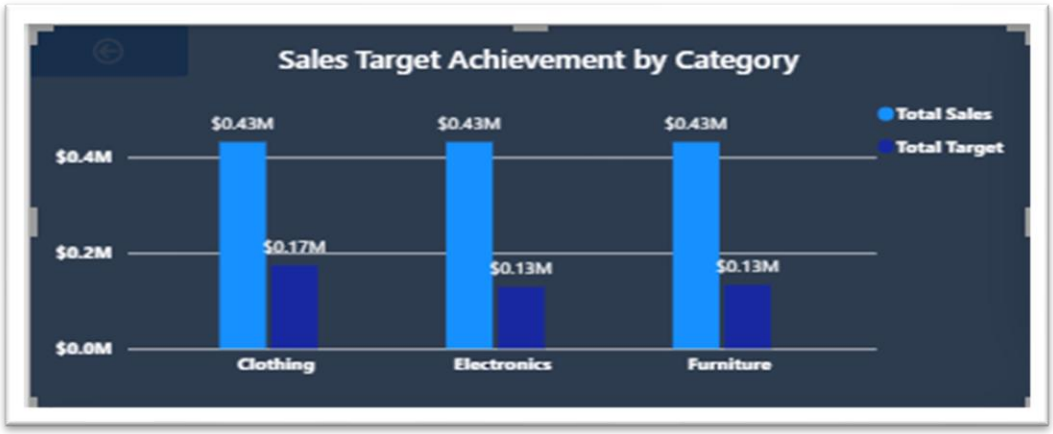
5. Build visualizations and dashboard:

- ***Sales Target Achievement by Category:***
  - ✓ Created a column chart by comparing the sales and target by category
  - ✓ For the axis used the Category and Total sales and Total Target

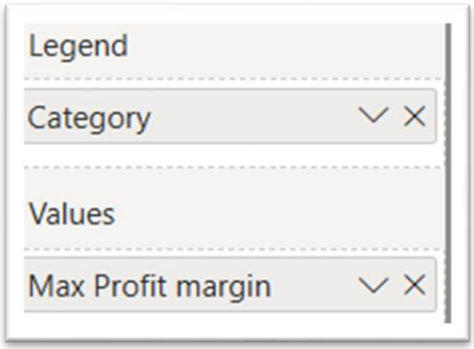


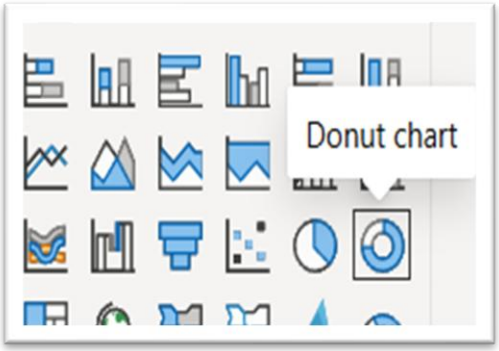


  - ✓ Customized the visual using **Format Visual → General** settings to enhance the appearance and clarity of the **Sales Target Achievement by Category** chart

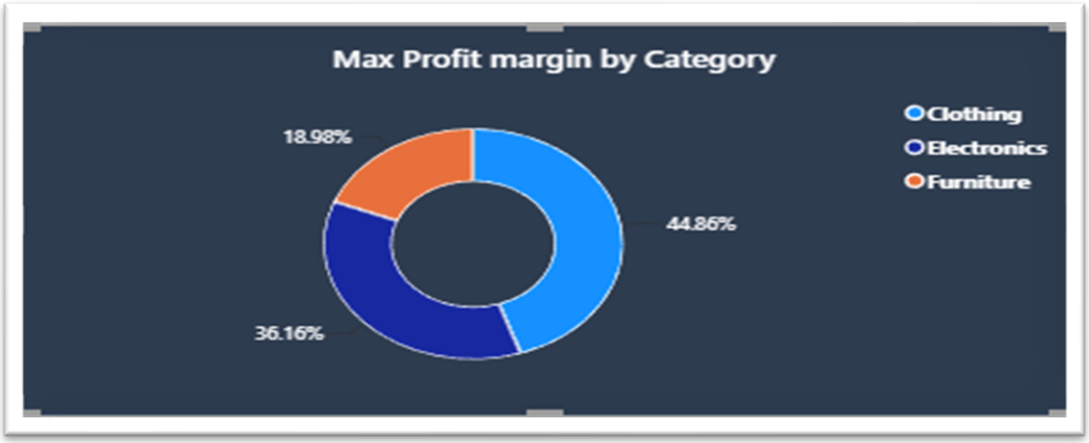


- ***<u>Max Profit Margin by Category:</u>***
  - ✓ Created an donut Chart by analysing the profit margin for each category

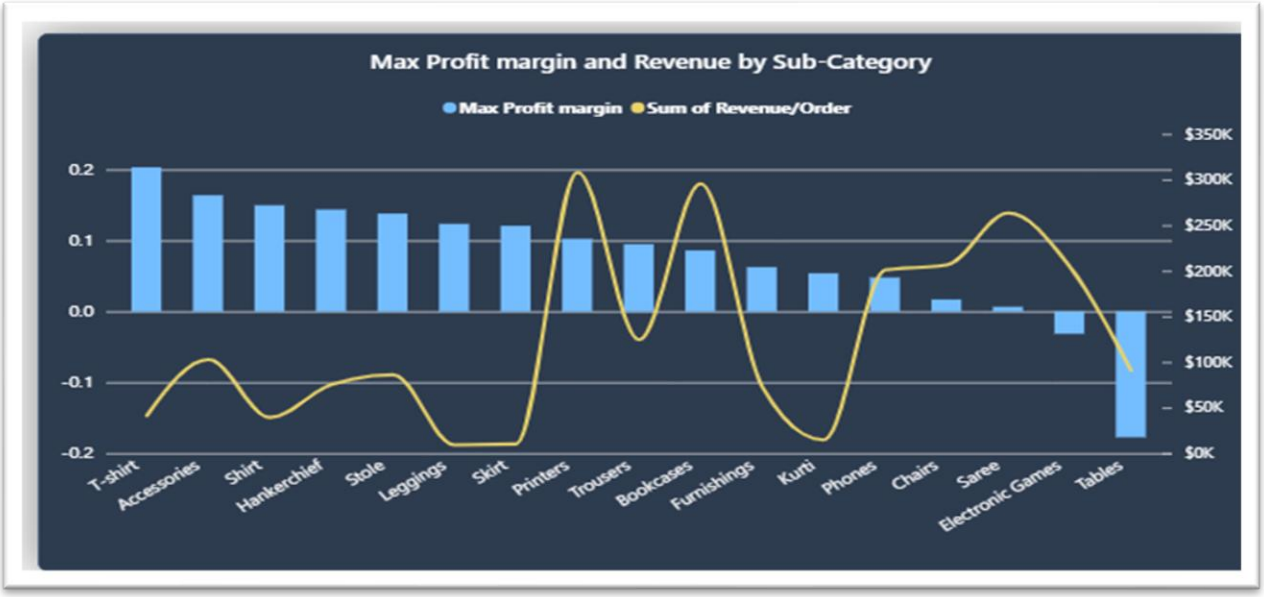




  - ✓ Using the **Format painter** Option applied all the format from the Column chart

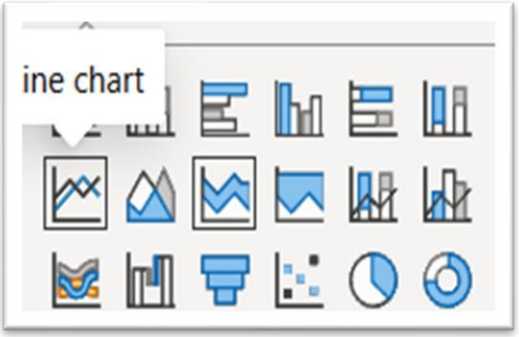


- ***<u>Max Profit Margin and revenue by SubCategory:</u>***
  - ✓ Created a line and Stacked column chart to represent the Max profit margin and revenue based on the subcategories



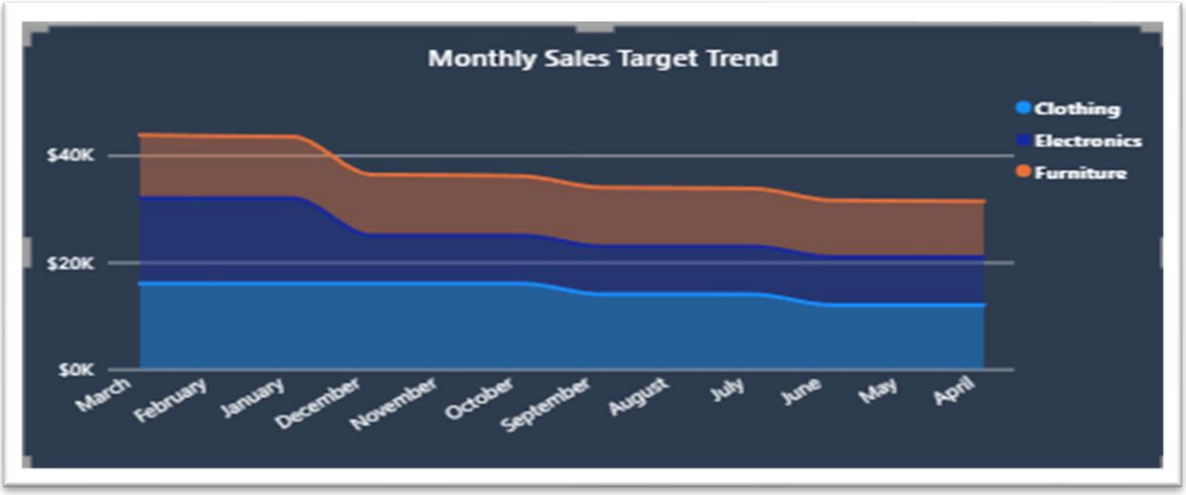
- ***<u>Monthly Sales Target Trend:</u>***

  - ✓ Created a line chart by comparing the targets for the month from sales target Table
  - ✓ Gave Axis as Month and Total target (Measure) separated by Category



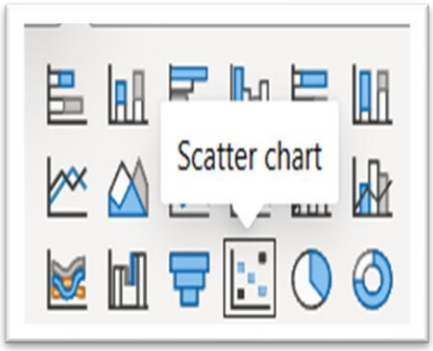


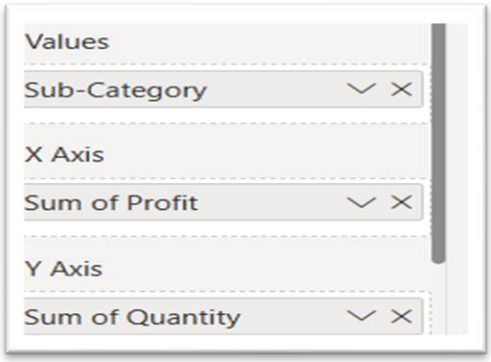
- ✓ Using the **Format painter** Option applied all the format from the Existing chart



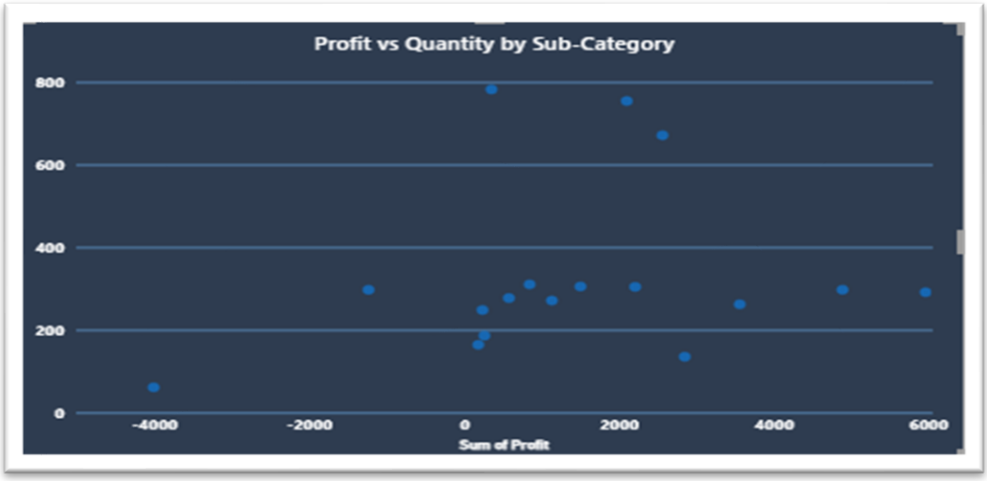
- ***Comparison of Profit and Quantity by Sub-Category:***

  - ✓ Compared the relationship between profit and quantity sold for different sub-categories using a scatter chart





  - ✓ Using the **Format painter** Option applied all the format from the Existing chart



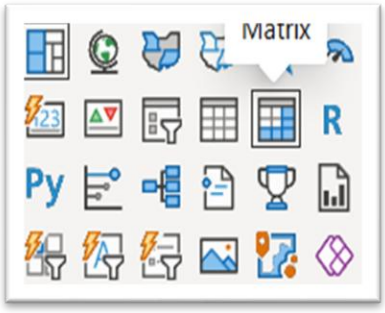
- ***Comparison of Total Sales Amount and Target:***
  - ✓ KPI cards provide a quick overview of ,
    - Total Profit
    - Total orders
    - Sales Achievement
    - Total Sales
    - Total Target
    - Progit Margin%

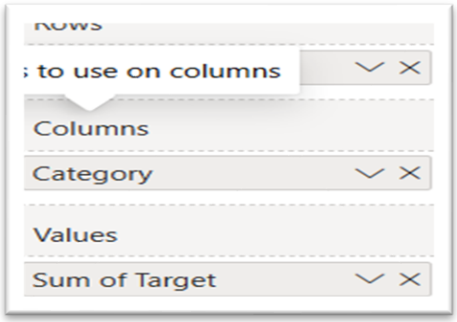


- ✓ Multirow card has visualized for each category in the Sales target table and minimum target



- ***Sales Performance Matrix:***
  - ✓ Built the matrix view to analyze how the Target distribution happened across category by month



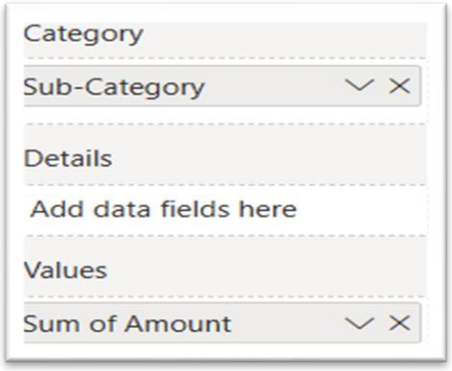


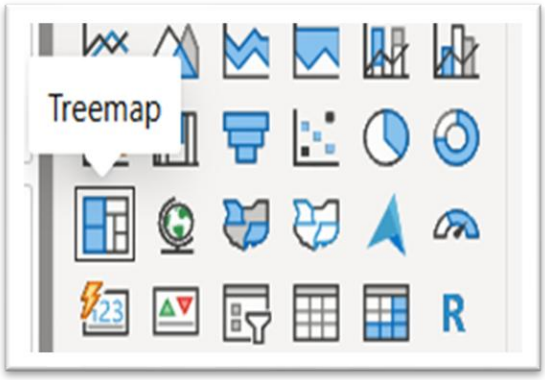
Sales Performance Matrix

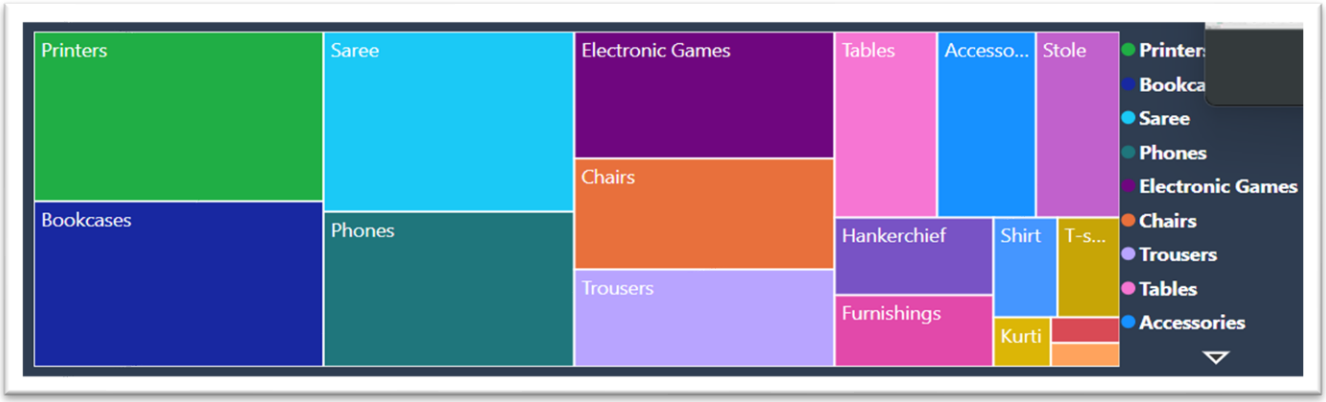
Month Name	Clothing	Electronics	Furniture	Total
April	$12,000.00	$9,000.00	$10,400.00	$31,400.00
August	$14,000.00	$9,000.00	$10,900.00	$33,900.00
December	$16,000.00	$9,000.00	$11,400.00	$36,400.00
February	$16,000.00	$16,000.00	$11,600.00	$43,600.00
January	$16,000.00	$16,000.00	$11,500.00	$43,500.00
July	$14,000.00	$9,000.00	$10,800.00	$33,800.00
June	$12,000.00	$9,000.00	$10,600.00	$31,600.00
March	$16,000.00	$16,000.00	$11,800.00	$43,800.00
May	$12,000.00	$9,000.00	$10,500.00	$31,500.00
November	$16,000.00	$9,000.00	$11,300.00	$36,300.00
October	$16,000.00	$9,000.00	$11,100.00	$36,100.00
September	$14,000.00	$9,000.00	$11,000.00	$34,000.00
Total	**$1,74,000.00**	**$1,29,000.00**	**$1,32,900.00**	**$4,35,900.00**

- ***Sales Distribution by Sub-Category:***

✓ To represent the sales distribution across different sub-categories using a treemap

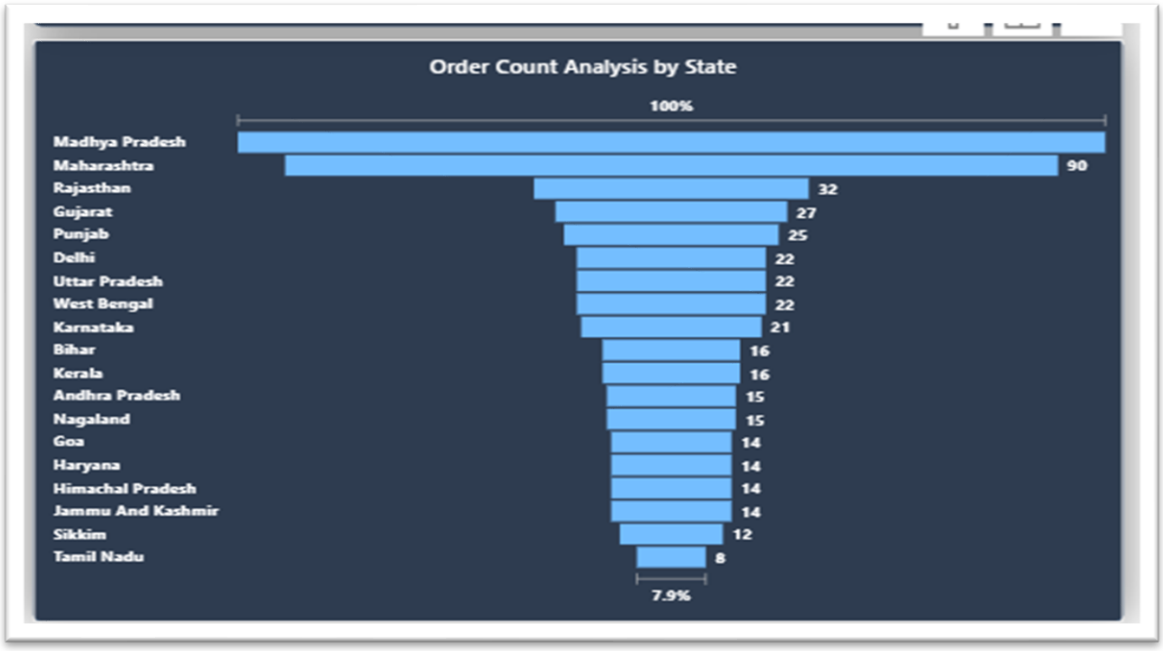






- ***Order Count Analysis by State:***

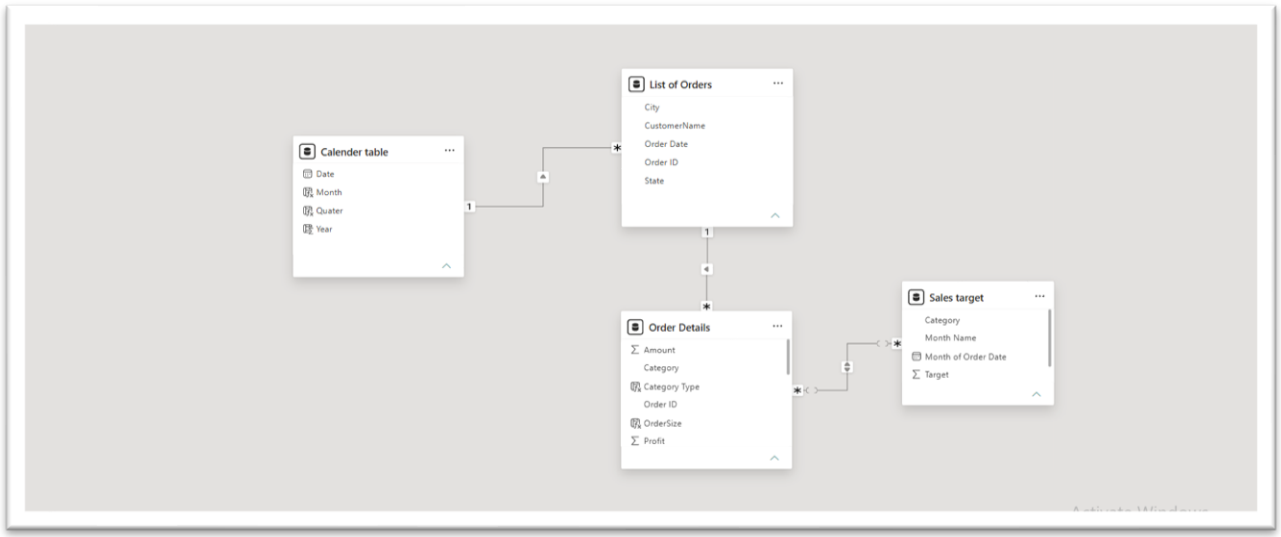
✓ Created a funnel chart to visualize the distribution of order counts across different states



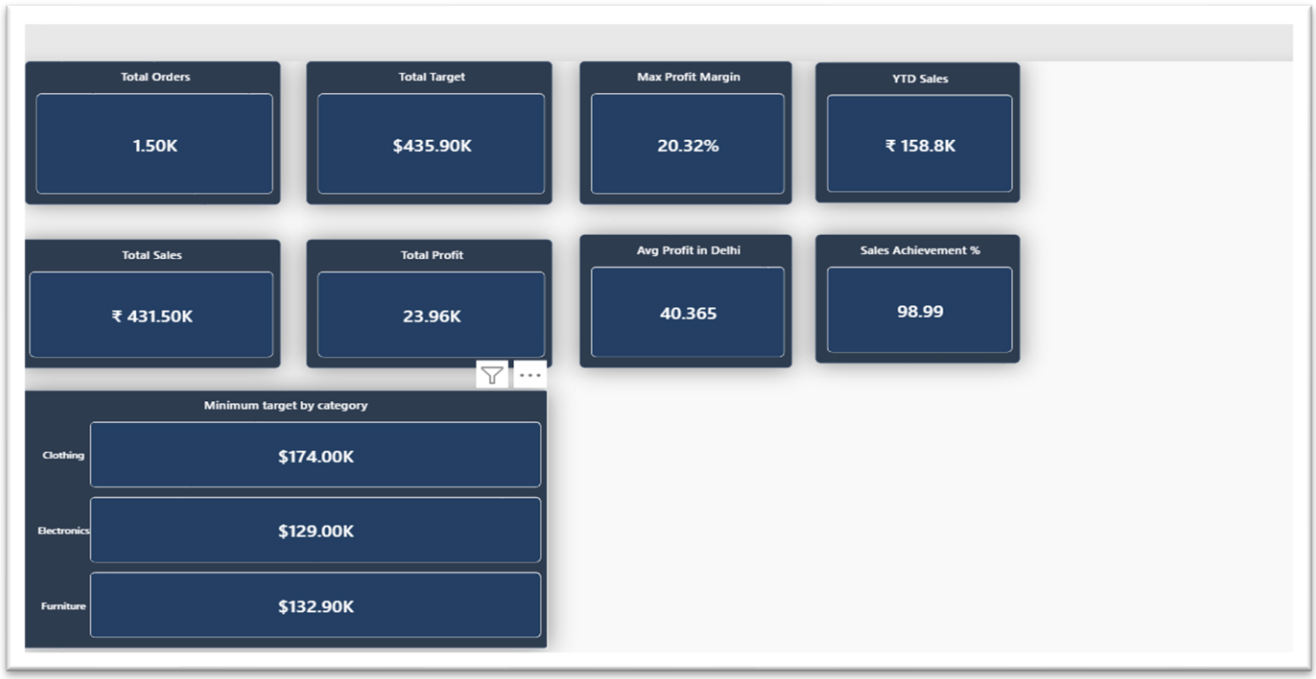
- ✓ Finally created an Dashboard by using the chart which are built using the tables and KPI to present good review of Ecommerce
- ✓ Applied Slicers from the Visual pane for SubCategory and Sales Category

6. Final output Screenshots :

- ✓ **<u>Data model:</u>**



✓ **KPI cards**



✓ **Charts and Slicers**







FINAL DASHBOARD:

